

Review

Poor treatment outcome and its determinants among patients with burn injuries in Ethiopia. A systematic review and meta-analysis

Mengistu Abebe Messelu^{a,*}, Baye Tsegaye Amlak^a, Tadesse Yirga Akalu^b,
Getnet Nibret Alemie^c, Salealign Amlaku Matebe^c, Mamaru Getie Fetene^d,
Bekele Getenet Tiruneh^e, Ashenafi Fekad Getahun^f, Makda Fekadie Tewelgne^g,
Tadesse Miretie Dessie^h, Yasab Leykunⁱ, Temesgen Ayenew^a

^a Department of Nursing, College of Medicine and Health Sciences, Debre Markos University, Debre Markos, Ethiopia

^b Department of Pediatrics and Child Health Nursing, College of Medicine and Health Sciences, Debre Markos University, Debre Markos, Ethiopia

^c Department of Pharmacy, College of Medicine and Health Sciences, Debre Markos University, Debre Markos, Ethiopia

^d Department of Midwifery, College of Medicine and Health Sciences, Debre Markos University, Debre Markos, Ethiopia

^e Department of Internal Medicine, School of Medicine, College of Medicine and Health Sciences, Debre Markos University, Debre Markos, Ethiopia

^f Department of Pediatrics and Child Health Nursing, School of Nursing, College of Medicine and Health Sciences, University of Gondar, Gondar, Ethiopia

^g Department of Health Informatics, Institute of Public Health, College of Medicine and Health Sciences, University of Gondar, Gondar, Ethiopia

^h Department of Public Health, College of Medicine and Health Sciences, Injibara University, Injibara, Ethiopia

ⁱ Department of Medical Laboratory, College of Medicine and Health Sciences, Debark University, Debark, Ethiopia

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ABSTRACT

Background: Burn continues to be a major public health problem, resulting in thousands of preventable deaths and disabilities each year. Therefore, this review and meta-analysis aimed to assess the national prevalence of poor treatment outcome and its determinants.

Methods: This systematic review was conducted according to the guidelines of Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) and has been registered with PROSPERO. A structured search of databases (Medline/PubMed, Google Scholar, CINAHL, EMBASE, HINARI, and Web of Science) was undertaken. The quality of the studies was assessed using the Joanna Briggs Institute (JBI) critical appraisal tool. A meta-analysis using a random-effects model was performed to estimate the pooled prevalence and associated factors. The I^2 statistics was used to assess the heterogeneity of studies, and to identify the possible causes of heterogeneity, subgroup analysis and meta-regression were used. Egger's test and funnel plots were used to assess publication bias.

Results: This systematic review and meta-analysis included eight studies involving a total of 9255 burn patients. The pooled national prevalence of poor treatment outcome among burn patients in Ethiopia was 20.1 % (95 %CI: 14.14, 26.06). Regional subgroup analysis showed that the pooled estimate of poor outcome was 13.12 %, 29.24 %, 26.2 %, 25.6 %, and 12.37 % in Tigray, Amhara, Oromia, Southern Nation Nationalities and People, and Addis Ababa, respectively. Moreover, subgroup analysis based on patients' age revealed that the pooled estimate of poor outcome among adults and children was 18.87 % and 21.58 %, respectively. This meta-analysis demonstrated that being a rural resident (OR = 3.24, 95 % CI: 1.66, 6.33), malnourished (OR = 3.17, 95 % CI: 1.93, 5.19), extent of burn > 10 % of TBSA (OR = 2.60, 95 % CI: 1.73, 3.90), and didn't get adequate fluid resuscitation (OR = 2.86; 95 % CI: 1.36, 6.00) were determinants of poor clinical outcome for burn patients.

Conclusion and recommendations: One in every five burn patients in Ethiopia had poor treatment outcome. Being a rural resident, malnourished, extent of burn > 10 %, and no getting adequate fluid resuscitation were significant

Abbreviations: CI, Confidence Interval; OR, Odds Ratio; PECO, Population, Exposure, Context, Outcome; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses; TBSA, Total Body Surface Area; WHO, World Health Organization.

* Corresponding author.

E-mail addresses: mengistuabebe@dmu.edu.et (M.A. Messelu), bayetsegaye07@gmail.com (B.T. Amlak), tadesseyirga680@gmail.com (T.Y. Akalu), getinetnibret@gmail.com (G.N. Alemie), salleamlak@gmail.com (S.A. Matebe), marugetie@gmail.com (M.G. Fetene), bekeleget12@gmail.com (B.G. Tiruneh), 98ashenafi@gmail.com (A.F. Getahun), makdafekadeal@gmail.com (M.F. Tewelgne), tadessemiretie19@gmail.com (T.M. Dessie), yasmagnaw1@gmail.com (Y. Leykun), teme31722@gmail.com (T. Ayenew).

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predictors of poor treatment outcome. Special attention should be given to burn injured patients from rural area, malnourished, and with larger TBSA.

1. Introduction

According to the World Health Organization (WHO), an estimated 180, 000 deaths every year are caused by burn, and the vast majority occur in low- and middle-income countries [1–3]. The WHO described burns as ‘forgotten global public health crises’ [4]. Burn is the fourth most common type of trauma worldwide, following traffic accidents, falls, and interpersonal violence [2]. Moreover, it is a global public health problem that is the leading cause of morbidity, prolonged hospitalization, disfigurement, and disability, often resulting in stigma and rejection [3,5,6]. Burns are also the leading cause of disability and disfigurement; fire-related burns alone are estimated to cause 10 million Disability Adjusted Life Years (DALY) [1,4,7].

A review from sub-Saharan Africa showed scalds were the commonest cause of thermal injuries, accounting for 59 % of all burns [8]. Furthermore, in sub-Saharan African countries, burn is a significant cause of death, with an average mortality rate of 17 %, and disproportionately affects those aged less than 15 years [8,9]. In Ethiopia, burns account for 1.5 % to 9 % of injuries in all age groups and 4 % to 15 % of injuries in pediatrics with an estimated mortality rate of 11.6 % [3].

Ethiopia has undergone increasing urbanization and development, with a concomitant increase in hazardous workplace conditions and electrical grid misuse [4]. As a result, burn injuries are on the rise and are fast becoming major health problems [4]. Thus, the Federal Ministry of Health (FMOH) takes the initiative with the stakeholders to expand the services and try to improve the survival outcome of burn injuries by emphasizing a multidisciplinary approach at all stages of management [4]. Despite remarkable efforts being made to improve the prevention, acute care, and rehabilitation of burn injuries, mortality and disability resulting from burn injuries remain high in Ethiopia.

According to different literature, the outcomes of burn injuries depend on the extent of injury, depth, age, sex, residency, time of arrival at health facility, condition at admission, getting appropriate fluid resuscitation, type of management given after admission, presence of comorbidity and co-existing injuries, and provision of pre-hospital care [8,10–16].

Several systematic reviews regarding burn injuries that include African countries have been employed, but they often identify the epidemiological pattern and results on the outcomes and its determinant factors are erratic [2,3,6,7]. Although burn continues to be a major public health problem in Ethiopia, resulting in thousands of preventable deaths and disabilities each year, there is a contradiction of results about the outcomes and factors associated with poor outcome among burn patients [12,14]. So far, no systematic review or *meta-analysis* has been conducted about the outcomes of burn injuries. Thus, this review and *meta-analysis* aimed to assess the pooled outcomes and factors associated with poor treatment outcome among burn patients in Ethiopia. The evidence generated from this systematic review and *meta-analysis* will be crucial for clinicians and policymakers to design, strengthen, and implement standards for burn care and treatment.

2. Methods

2.1. Study protocol and registration

The protocol for this review has been pre-registered with the International Prospective Register of Systematic Reviews (PROSPERO) under the unique identification number (UIN): review registry CRD42023502561. This systematic review and *meta-analysis* has been reported in line with the preferred Reporting Item for Systematic Review and *Meta-analysis* (PRISMA) guidelines [17].

2.2. Searching strategy

We conducted a comprehensive literature search across different electronic databases, such as PubMed, Scopus, Google Scholar, CINAHL, EMBASE, HINARI, and Web of Science, from inception to January 10, 2024. There were no limits to publication year or study design, but an English-language filter was used. Additionally, to find unpublished studies and grey literature, we searched Google Scholar from the earliest to January 10, 2024; all published literature was retrieved, critically evaluated, and assessed for inclusion in this review. Conference presentations, committee reports, and continuing research are not included in this systematic review because they do not fully depict the results, and the results may change completely when they are published. The PECO (Population, Exposure, Context, Outcome) search strategy was used. The electronic databases were searched with keyword searching and using the Medical Subject Heading (MeSH) terms for each selected PECO component. Search terms used to access studies are the following: ‘burn’, ‘burns’, ‘thermal injury’, ‘outcome’, ‘outcomes’, ‘poor outcome’, ‘unfavorable outcome’, ‘outcome with complications’, ‘Ethiopia’, ‘Federal Democratic Republic of Ethiopia’, ‘associated factors’. These search strategies were combined using ‘AND’ and/or ‘OR’ Boolean operators (Supplementary Table 1).

2.3. Inclusion and exclusion criteria

Observational studies (prospective and retrospective cohort, cross-sectional, and case control) that reported the prevalence of outcome, conducted in Ethiopia, and published in English were included in this review. Articles that did not provide full access and didn’t report the treatment outcome were excluded.

2.4. Outcomes

This review and *meta-analysis* primarily focus on determining the pooled prevalence of poor treatment outcome among patients with burn injuries in Ethiopia, expressed as a percentage with frequency. The calculation entails dividing the total number of patients who had poor treatment outcome by the overall number of patients who sustained burn injuries, multiplied by 100. Poor treatment outcome included those burn patients who died or discharged with complications such as disability, disfigurement, or contracture. The authors utilized the Odds Ratio (OR) as an outcome measure to pinpoint determinants of poor outcome. Factors reported in at least two studies were considered for pooled analysis.

2.5. Data extraction and data quality assessment

The articles identified in the literature search were first screened by title and abstracts for inclusion in the systematic review by independent reviewers (M.A.M., T.A., and B.T.A.). Those studies that passed the title and abstract screening processes were eligible for full-text review. The independent reviewers (M.A.M., T.A., B.T.A., S.A.M., and G.N.A.) evaluated the full text of the eligible studies for inclusion in the final analysis. The disagreement was handled based on the inclusion and exclusion criteria, and the third reviewer (T.Y.A.) overcame the disagreements between the first two.

Microsoft Excel 2019 was used to extract data including the name of the first author, year of publication, region of the study, prevalence of poor outcome, sample size, study design, and population age group. Two independent reviewers (M.A.M. and T.A.) extracted and cross-checked the extracted data for potential variations, and the discrepancies were

handled by re-evaluating the full text. A third review (B.T.A.) examined the extracted data to ensure its accuracy and identify any errors.

The Joanna Briggs Institute (JBI) quality assessment tool [18] for prevalence studies was used to assess the quality of the included studies. The criteria were: 1) sample representativeness; 2) sample size adequacy; 3) using valid measurements; 4) using appropriate statistical analysis methods; and 5) response rates. Studies that scored greater than or equal to 50 % of the assessment criteria were considered to have a low risk of bias.

2.6. Data analysis

Endnote X-8 reference manager software was used to manage the selection process. STATA version 17 was used for data analysis. Descriptive statistics were employed to summarize the extracted data. Potential publication bias was checked by the funnel plot [19] and Egger's test [20]. The heterogeneity of articles was assessed using I^2 statistics [21]. Depending on the heterogeneity of the included studies, pooled estimates were calculated for key outcomes, such as poor outcome and discharge with complications. DerSimonian and Laird's random effects model was used to estimate the pooled prevalence with a 95 % CI. Subgroup analysis was done using region, type of population, and age group. Furthermore, the sensitivity test was done to check the effect of individual studies on the pooled estimate.

3. Results

3.1. Article selection

Using the PRISMA flow diagram, a total of 323 potentially eligible studies were identified. After removing duplications and conducting a full-text review, 13 studies were retrieved. Finally, eight studies remained in this systematic review and Meta-analysis. Three articles [14,22,23] that didn't report the outcome were subsequently excluded (Fig. 1).

3.2. Study characteristics

Studies included in this review and meta-analysis were conducted in

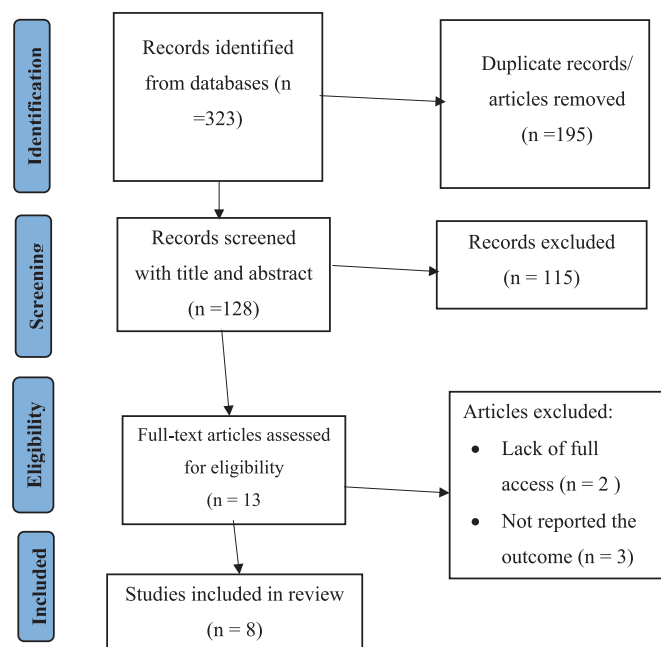


Fig. 1. PRISMA flow-chart to represent the selection process of articles in Ethiopia.

Ethiopia and published in indexed journals. All included studies were cross-sectional, and conducted in Addis Ababa [13,15], Amhara [16,24], Oromia [25], South Nation and Nationalities [12], and Tigray [11,26]. A total of 9255 patients with burn injuries were included, and the sample size ranged from 102 to 7309 (Table 1).

4. Prevalence of poor treatment outcome

As mentioned in Fig. 2, the prevalence of poor outcome among patients with burn injuries in Ethiopia was 20.1 % (95 %CI: 14.14, 26.06) (Fig. 2).

4.1. Subgroup analysis

The subgroup analysis using regions of the country showed that the highest proportion of burn patients with poor treatment outcome was observed in the Amhara region (29.94 %), followed by the Oromia region (26.20 %) (Fig. 3).

Similarly, another subgroup analysis using the age of the study population revealed that the proportion of poor treatment outcome among children is higher (21.58 %) as compared to all age groups (18.87 %) (Fig. 4).

4.2. Publication bias evaluation

The presence of publication bias was evaluated using Egger's ($p = 0.000$) and Begg's tests ($p = 0.013$), which showed that there is a significant publication bias. In addition to the statistical tests, a funnel plot was used, and it looks asymmetrical. The vertical line indicates the effect size, whereas the diagonal lines indicate the precision of individual studies with a 95 % CI (Fig. 5).

4.3. Investigation of heterogeneity

A meta-regression analysis using the sample size and publication year was conducted to identify the possible source of heterogeneity. However, none of these factors showed significant heterogeneity. Therefore, the source of heterogeneity remains unknown (Table 2).

5. Factors associated with poor outcome of burn patients

5.1. Residence

The odds of having poor outcome among burn patients who reside in rural areas are 3.2 times (OR = 3.24, 95 % CI: 1.66, 6.33) higher as compared to those who reside in urban areas (Fig. 6).

5.2. Extent of burn

The pooled effect of three studies was reported on the association between extent of burn and poor outcome of burn patients. This meta-analysis study revealed that those who had > 10 % of TBSA had 2.6 times (OR = 2.60, 95 % CI: 1.73, 3.90) higher odds of having a poor outcome as compared to their counterparts. There is no observed heterogeneity ($I^2 = 0.0$ %) (Fig. 7).

5.3. Nutritional status

The pooled effect of two studies was reported on the association between nutritional status and poor outcome. This meta-analysis study revealed that those who were malnourished had 3.2 times higher odds of having a poor outcome as compared to those who were well-nourished (OR = 3.17, 95 % CI: 1.93, 5.19). There is low observed heterogeneity ($I^2 = 15.0$ %) (Fig. 8).

Table 1
General characteristics of the included studies.

Author	Publication year	Region	Age group	Sample size	Poor outcome	Study design	Quality
Dessie et al [16]	2023	Amahra	Children	250	27.6	Cross sectional	Low risk
Mulatu et al [15]	2022	AA	All	241	14.9	Cross sectional	Low risk
Nega et al [26]	2002	Tigray	All	7309	9.4	Cross sectional	Low risk
Alemayehu et al [11]	2020	Tigray	Children	382	17.3	Cross sectional	Low risk
Mamo et al [13]	2023	AA	Children	256	10.2	Cross sectional	Low risk
Biru and Mekonen [12]	2021	SNNP	All	395	25.6	Cross sectional	Low risk
Bekri et al [25]	2020	Oromia	All	420	26.2	Cross sectional	Low risk
Gessesse et al [24]	2020	Amhara	Children	102	33.3	Cross sectional	Low risk

AA; Addis Ababa, SNNP; Southern Nation Nationalities and People of Ethiopia.

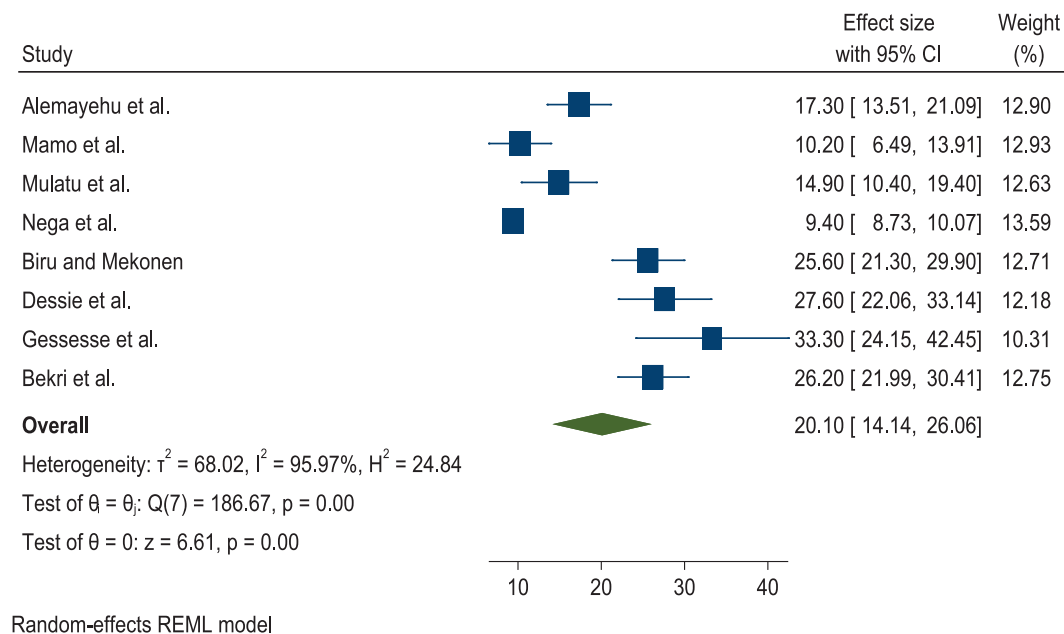


Fig. 2. Forest plot to depict the pooled prevalence of poor outcome of patients with burn injuries in Ethiopia.

5.4. Fluid resuscitation

According to the pooled estimates of three articles, the outcomes of patients with burn injuries were significantly associated with fluid resuscitation. Those who didn't get adequate fluid resuscitation were 2.9 times (OR = 2.86; 95 % CI: 1.36, 6.00) more likely to have a poor treatment outcome as compared to those who resuscitated adequately (Fig. 9).

5.5. Sensitivity analysis

The result of sensitivity analysis using the random effect model showed that no single study unduly affected the pooled prevalence of poor treatment outcome among burn patients (Table 3).

6. Discussion

Burn injuries are the most devastating and disabling trauma to human being, and it remains a serious threat to the well-being of people in all age groups, which has major aesthetical, physiological, and functional sequels. This systematic review and meta-analysis estimated the national prevalence of poor treatment outcome and its associated factors among burn patients in Ethiopia.

This review and meta-analysis found that the national-level prevalence of poor outcome among patients with burn injuries in Ethiopia was 20.1 %. This result is comparable with the study done in Sudan [27], which revealed that about 20 % of burn patients are discharged with

complications. This pooled prevalence of poor treatment outcome is lower than a systematic review and meta-analysis conducted in Cameroon [28], which reported that the proportion of mortality among patients with burn injuries was 29 %. This might be due to the heterogeneity of the study population in terms of socio-demographic characteristics, the differences in study settings, mechanisms of injury, severity of burn injuries, and operational definitions of outcomes [29].

However, it is higher compared with the study conducted in Tanzania (1.7 %) [30]. The possible reason for this discrepancy might be due to the fact that most patients had TBSA < 20 %, which can be a contributing factor for low mortality in Tanzania. It is also higher than the reviews done in Kenya [31] and Iran [32], which reported that the proportion of poor treatment outcomes among burn patients was 9.3 % and 14 %, respectively. This could be due to the differences in standard of care, availability of trained health professionals, and well-equipped burn centers in developed countries [22]. Thus, the government of Ethiopia, in collaboration with other stakeholders, needs to design strategies to reduce the poor treatment outcomes of burn patients.

This review also found that the proportion of poor treatment outcome among pediatric burn patients was higher as compared with all age groups. This might be due to the fact that younger children are easily injured because of their curiosity and their explorative manners about what surrounds them [33]. In addition, younger children have low physiologic thresholds to resist the effect of burn injury [34]. On the contrary, the study conducted in Canada [35], Netherlands [36], and USA [37] reported that the risk of poor outcome increases linearly as age increases. Thus, remarkable attention ought to be paid to children, and it

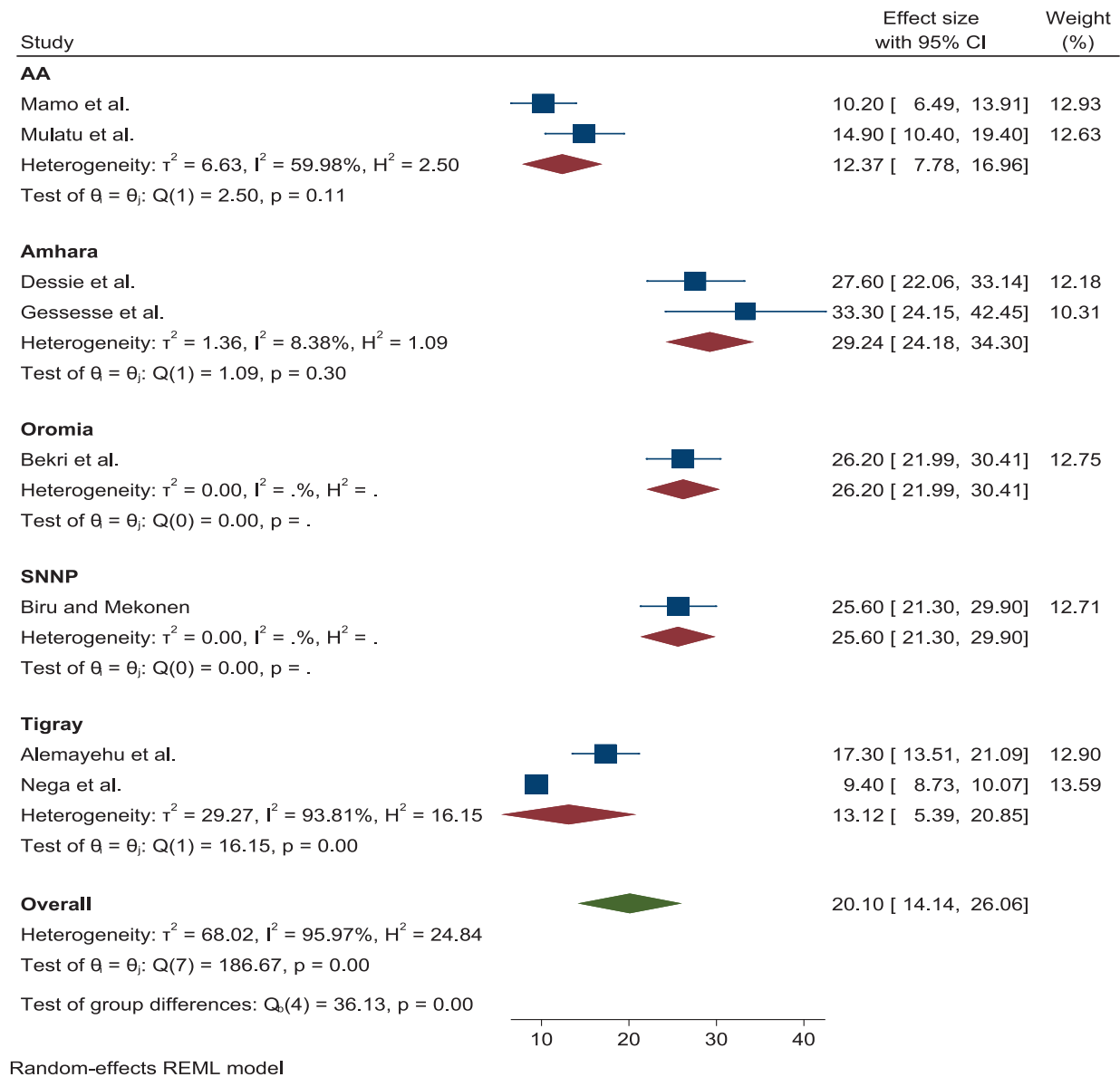


Fig. 3. Forest plot to depict the pooled prevalence of poor outcome among burn patients by regions in Ethiopia.

is also important to limit their access to hot liquids or electrical outlets.

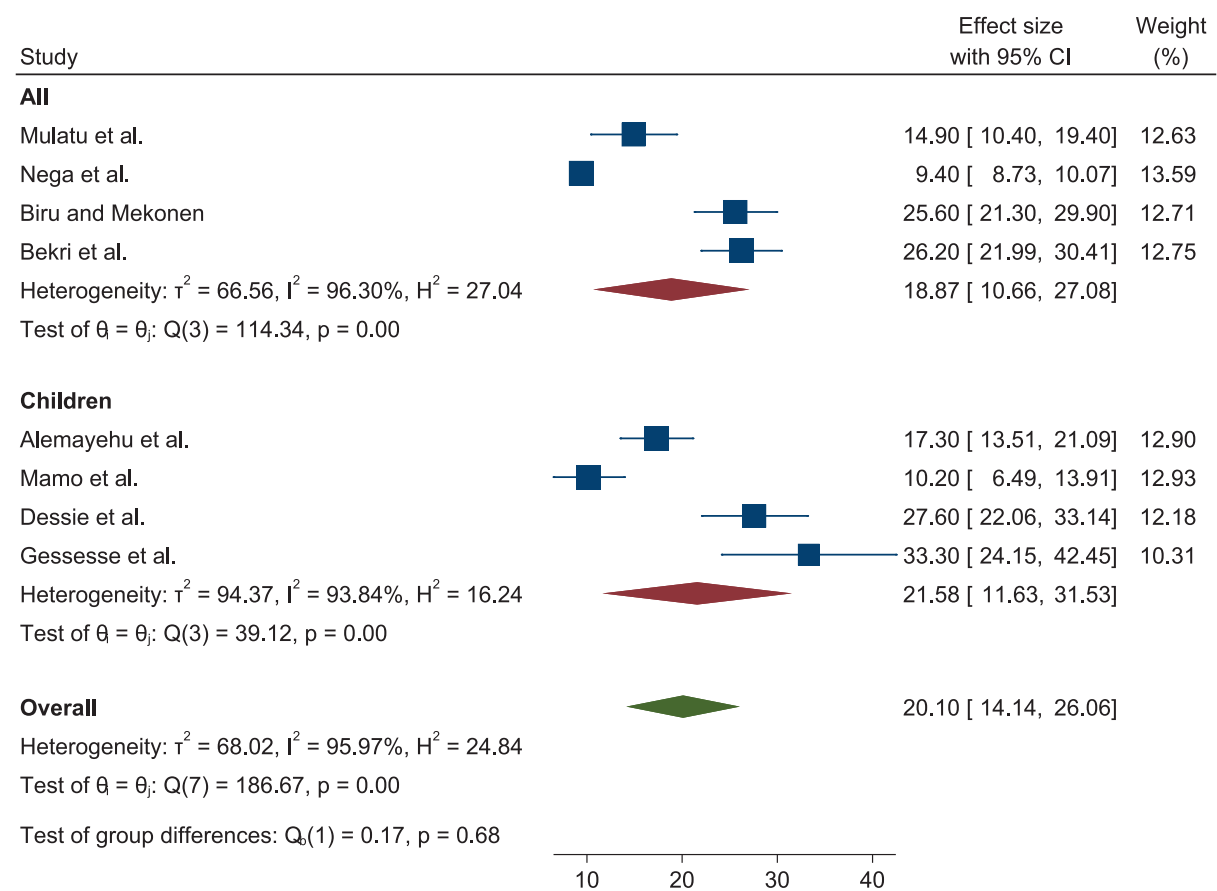
The odds of having a poor outcome among burn patients who reside in rural areas are 3.2 times higher as compared to those who reside in urban areas. This might be due to the fact that increased use of open flames for heating and lighting in rural areas increases the risk of poor outcome. Areas of burn accidents are the most critical factors that influence in providing safety measures to prevent or decrease further damage caused by burn injuries [32]. Furthermore, this could be explained by the differences in early accessing health facilities to get immediate interventions, such as resuscitation [38].

This meta-analysis study revealed that those who had > 10 % of TBSA had 2.6 times higher odds of having a poor outcome as compared to their counterparts. This finding is consistent with the reviews done in Europe [38], Cameroon [28], Kenya [31], Iran[39], and Sweden [40], where an increased percentage of total body surface area burned [TBSA] increases the risk of poor treatment outcome. This is explained by the scientific fact that those who have a larger body surface area of burn are prone to complex fluid shifts, including system-wide extravasation of fluids into unburned tissues [41]. The higher the extent of burn, the greater the risk of the subsequent burn wound and systemic infections, which results in

intravascular hypovolemia, which can produce ischemia, lactic acidosis, and cardiovascular collapse, which increases the risk of mortality and complications [39].

Moreover, the odds of having a poor outcome among burn patients who were malnourished were 3.2 times higher as compared to those who were well-nourished. This finding is consistent with the studies conducted in Canada [42] and Malawi [43], which explained that pre-existing malnutrition in burn patients interferes with tissue regeneration and the ability to recover, which increases the odds of a poor outcome. It is well documented that the hypermetabolic response, or an increased basal metabolic rate, associated with burn injury requires a high calorie intake to allow optimal healing and outcome [44]. This is also supported by the scientific rationale that burn patients suffering from severe fluid loss, anemia, infection, increased catabolism, and/or decreased intake exhibit poor clinical outcome, which in turn eventually lead to death [23].

According to the pooled estimates of three articles [11,16,25], outcomes of patients with burn injuries were significantly associated with fluid resuscitation. Those who didn't get adequate fluid resuscitation were 2.9 times more likely to have a poor treatment outcome as



Random-effects REML model

Fig. 4. Forest plot to depict the pooled prevalence of poor outcome among burn patients by age groups in Ethiopia.

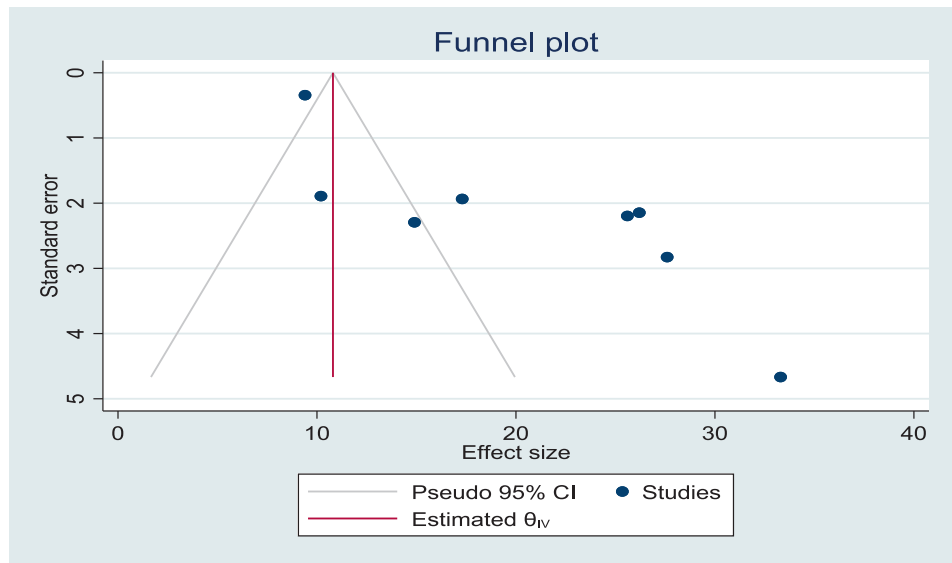


Fig. 5. Funnel plot to illustrate the presence of publication bias.

compared to those who were resuscitated successfully. This could be explained by the fact that hypotension reduces vital organ perfusion, which leads to multi-organ failure such as renal failure, heart failure, and brain death [45]. Therefore, the survival rates of burn patients directly depend on adequate fluid resuscitation because the most devastating effect after significant burn injuries is significant fluid loss.

Moreover, it is a widely accepted fact that severe fluid loss is the greatest problem following major burn injuries [31,46]. Therefore, burn patients need fluid resuscitation to maintain tissue perfusion and to halt hypovolemia that can occur due to complex fluid shifts within the body tissues [46].

Table 2
Meta-regression of poor treatment outcome of burn patients in Ethiopia with publication year and sample size to detect heterogeneity.

Source of heterogeneity	Coefficients	Standard error	p-value
Publication year	−0.0102416	0.006757	0.130
Sample size	−3.021152	2.81777	0.205

6.1. Limitations of the study

One of the limitations of this systematic review and meta-analysis was that only articles published in English-language were included, so any relevant articles published in other languages were not incorporated. Furthermore, the studies included in this systematic review and meta-analysis were conducted in only five regions of the country, which may restrict the representativeness of the findings.

6.2. Implications for nursing practice

Most patients with burn injuries experience significant fluid loss, necessitating prompt and effective fluid resuscitation to maintain tissue perfusion and prevent hypovolemic shock. Nurses play a critical role in initiating and monitoring fluid therapy, particularly within the first 24 h of admission when mortality risk is highest. This includes regular assessment of urine output, vital signs, and clinical indicators of hypovolemia to ensure adequate resuscitation.

Malnutrition, a prevalent and modifiable risk factor, impairs wound healing and increases the risk of infection. Nurses should incorporate routine nutritional screening into both initial and ongoing patient assessments. In resource-limited settings, they are also key in educating patients and caregivers on the importance of consuming affordable, locally available protein-rich foods to promote recovery.

Furthermore, nurses are responsible for prioritizing care for patients with extensive burns. This includes initiating early fluid resuscitation, administering pain management, implementing infection prevention

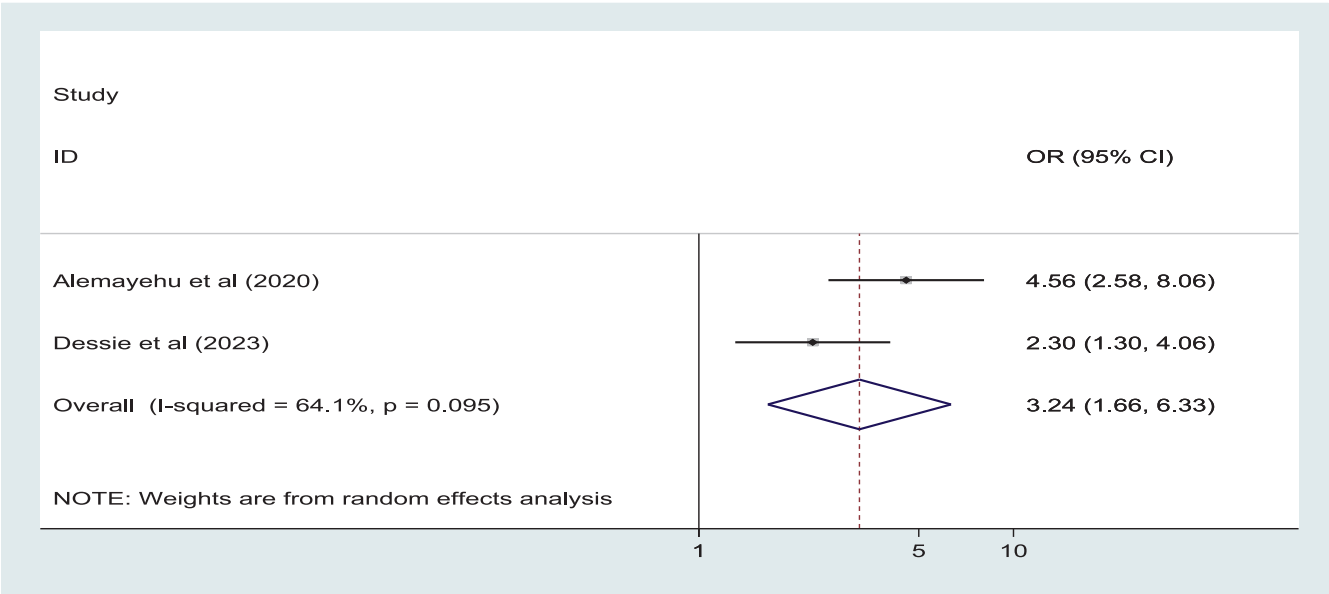


Fig. 6. The association between residence and treatment outcome of burn patients in Ethiopia.

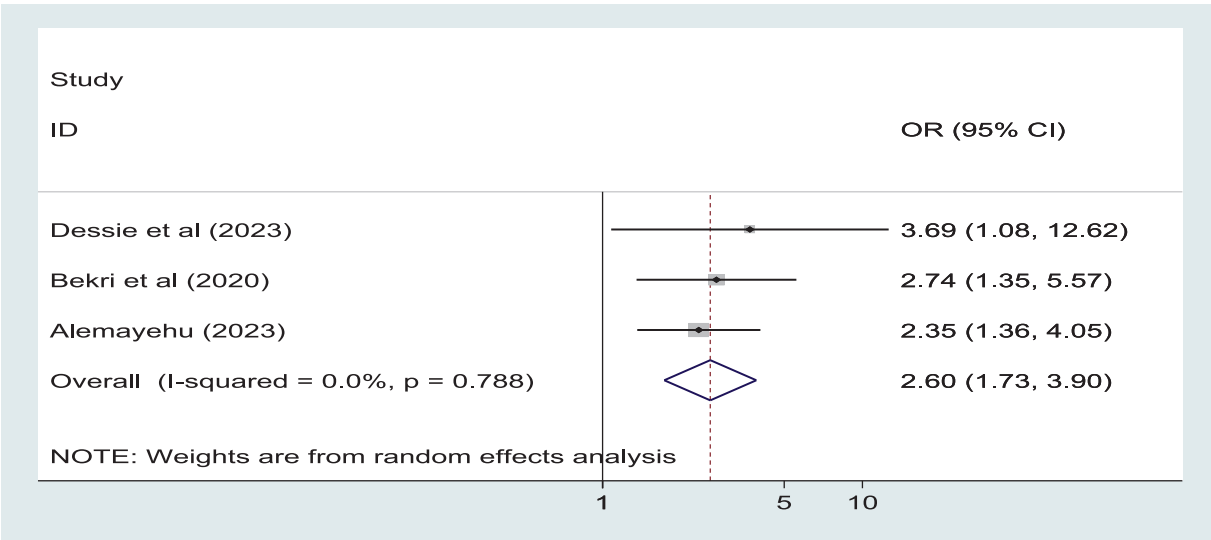


Fig. 7. The association between extent of burn and poor treatment outcome of burn patients in Ethiopia.

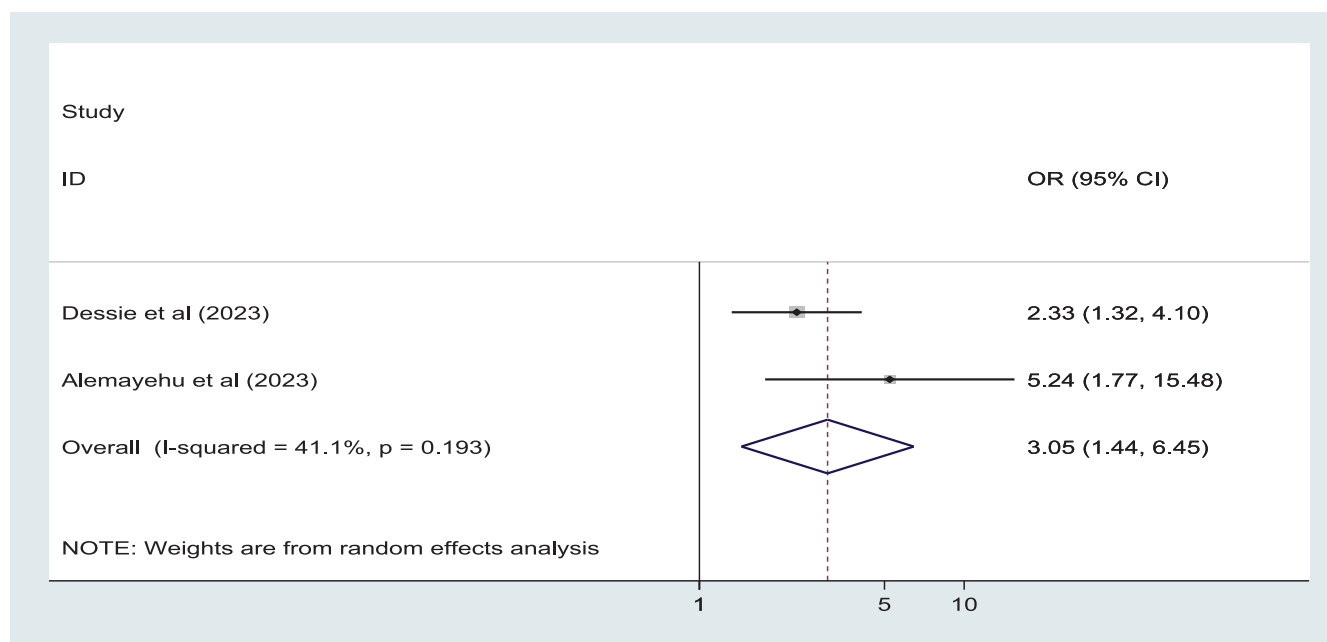


Fig. 8. The association between nutritional status and poor treatment outcome of burn patients in Ethiopia.

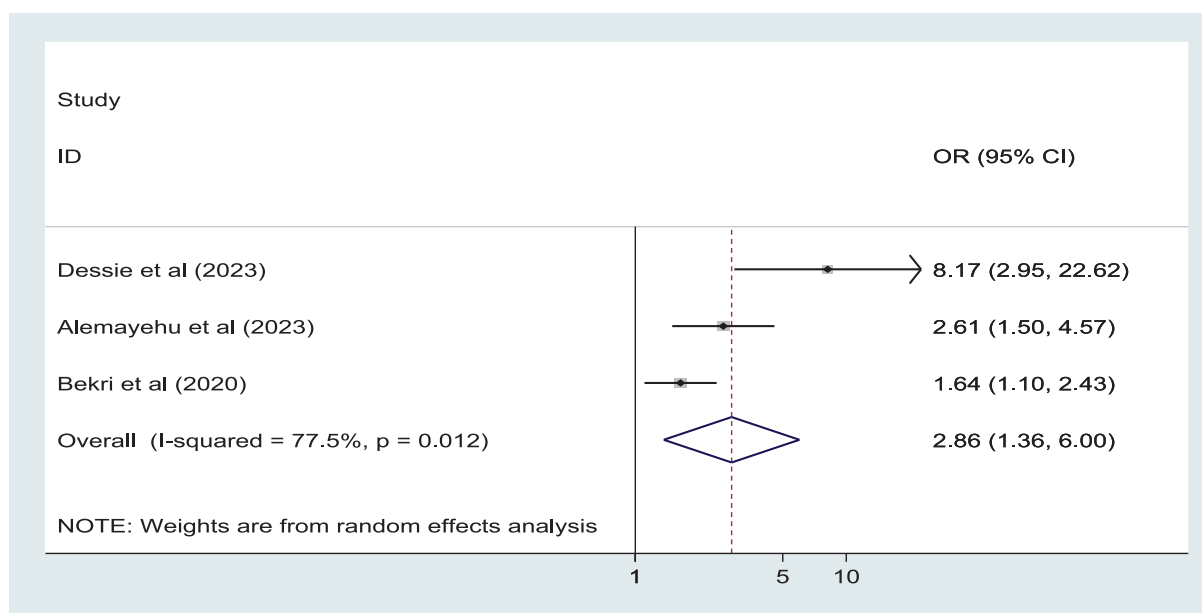


Fig. 9. The pooled estimate of the association between fluid resuscitation and poor treatment outcome among burn patients in Ethiopia.

Table 3

The sensitivity analysis to estimate the effect of a single study on the combined prevalence of the poor treatment outcome among burn patients in Ethiopia.

Study omitted	Prevalence (95 % CI)
Alemayehu et al, 2020	20.60 (13.52, 27.69)
Mamo et al, 2023	21.68 (14.33, 29.02)
Mulatu et al, 2022	20.94 (13.91, 27.96)
Nega et al, 2002	21.69 (16.01, 27.38)
Biru and Mekonnen, 2021	19.27 (13.06, 25.49)
Dessie et al, 2023	19.04 (12.81, 25.26)
Gessesse et al, 2020	18.58 (12.36, 24.79)
Bekri et al, 2020	19.16 (13.06, 25.27)
Combined	19.16 (13.06, 25.27)

measures, providing appropriate wound care, and advocating for transfer to higher-level care facilities when necessary.

7. Conclusion

This systematic review and meta-analysis found that the national magnitude of poor outcome among patients with burn injuries in Ethiopia was high, putting a high healthcare burden on the country, which requires collaboration between all stakeholders to improve outcome. Being a rural resident, malnourished, having an extent of burn > 10 % of TBSA, and not getting adequate fluid resuscitation were significant determinant factors for poor treatment outcome. Special attention should be given to burn injured patients from rural area, malnourished, and with larger TBSA. Furthermore, the federal ministry

of health and clinicians should focus on the provision of integrated, strengthened, sustainable, and standard burn care services to improve the outcomes of burn patients in our country.

Consent for publication

Not applicable.

Ethical approval and consent to participate

Because it is a systematic review and *meta-analysis*, ethical approval is not applicable.

Authors' contributions

MAM, BTA and TA involved in developing the protocol and study design, study selection, data extraction, statistical analysis, and editing of the final manuscript draft. GNA, TYA and SAM: involved in quality assessment, statistical analysis, and manuscript revision. TMD, AFG, YL, MGF, and BGT: study selection and extraction; took part in drafting the manuscript or revising it critically for important intellectual content; agreed to submit it to the current journal; gave final approval of the version to be published; and agreed to be accountable for all aspects of the work.

Availability of data and materials

All data supporting the findings of this study are available within this paper and its [supplementary information](#).

CRediT authorship contribution statement

Mengistu Abebe Messelu: Writing – review & editing, Writing – original draft, Software, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Baye Tsegaye Amlak:** Writing – review & editing, Supervision, Software. **Tadesse Yirga Akalu:** Validation, Software, Formal analysis. **Getnet Nibret Alemie:** Writing – review & editing, Validation, Supervision, Software. **Salegn Amlaku Matebe:** Writing – review & editing, Supervision, Formal analysis. **Mamaru Getie Fetene:** Writing – review & editing, Supervision, Software. **Bekele Getenet Tiruneh:** Investigation, Formal analysis, Data curation. **Ashenafi Fekad Getahun:** Writing – review & editing, Validation, Supervision. **Makda Fekadie Tewelgne:** Methodology, Formal analysis, Data curation. **Tadesse Miretie Dessie:** Methodology, Formal analysis, Data curation. **Yasab Leykun:** Supervision, Methodology, Formal analysis. **Temesgen Ayenew:** Writing – review & editing, Validation, Supervision, Software, Methodology, Formal analysis.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ienj.2025.101676>.

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